

# A conditional $2^{-76}$ exponent saving by tuning direct axis routing

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## Abstract

Combining our direct nonadjacent routing audit with the previously audited variable stopping threshold gives  $\kappa = 2^{-76}$  in  $O(n(\log n)^{1-\kappa})$ , conditional on OpenAI's underlying algorithmic interfaces. This quadruples our previous supplied exponent saving without changing the finite network or routing schedule. We give a transferable parameter recipe and a simple upper bound on further tuning under the current constraints; this is not a runtime speedup or an optimality claim.

## 1 Scope and parameter recipe

The source is OpenAI's *Integer multiplication below  $n \log n$* , pinned at

`adc7f1241b42e322a6451854ab7e4b4c146bf78a`.

We retain its fixed finite-alphabet, fixed finite number of one-dimensional tapes model. Our preceding notes establish conditional substitutions for the  $h = 46$  network, variable stopping exponent, and direct nonadjacent axis routing. Their underlying network and machine interfaces remain assumptions here.

**Proposition 1.** *Suppose compatible primitives support  $\tau = \sigma = 1 - a$ , with rational  $0 < a < 1/16$ , and the guard exponent remains  $C_1 = 20$ . Set*

$$\begin{aligned} \epsilon &= \frac{1}{21}, & \beta &= \frac{9}{10}, & c &= \frac{9a}{10}, & \delta &= \frac{1}{16}, \\ \lambda &= 1 - \frac{19a^2}{20}, & \lambda' &= 1 - \frac{9a^2}{10}. \end{aligned}$$

*Under the nonadjacent routing accounting, all recorded parameter constraints hold, and the smallest of the seven exponent margins is  $G = 3a^2/70$ . Consequently every fixed rational  $0 < \kappa < G$  is supported by the conditional assembly argument.*

*Proof.* The packed overhead and leaf inequalities are

$$\tau(1 + c/\beta) = 1 - a^2 < \lambda < \lambda' < 1, \quad \sigma + \beta(1 - \sigma) = 1 - a/10 < \lambda'.$$

The last comparison follows from  $a < 1/9$ . Also  $\lambda > \max\{\tau, \sigma\}$ ,  $\epsilon < 1/12$ ,  $\epsilon C_1 = 20/21 < 1$ , and  $0 < \delta < 1/8$ . The seven margins are

$$\begin{aligned} g_1 &= 1 - \frac{1}{21}(1 + 9a/10), & g_2 &= \epsilon ca = \frac{3a^2}{70}, \\ g_3 &= \epsilon(1 - \lambda') = \frac{3a^2}{70}, & g_4 &= a(1 - \epsilon) = \frac{20a}{21}, \\ g_5 &= \frac{1}{4} - \delta - \frac{3}{2}\epsilon = \frac{13}{112}, & g_6 &= 1 - \delta - \epsilon = \frac{299}{336}, & g_7 &= \frac{1}{21}. \end{aligned}$$

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\*Prepared with assistance from OpenAI Codex. This is not independent review or endorsement by OpenAI.



Hence  $c < \beta a / (1 - a) < a / (1 - a)$ , so every witness under these conditions obeys

$$\kappa < g_2 = \epsilon ca < \frac{a^2}{20(1 - a)} \approx 1.62000000003 \cdot 10^{-23} < 2^{-75}.$$

This necessary upper bound is not claimed sharp. It fixes the primitive exponent and current guard/layer constraints; it is not a lower bound on integer multiplication or a barrier to improved primitives or accounting. It explains why another parameter-only search cannot reach the 60s or 50s.

## 4 Reproducibility

The companion repository includes these artifacts:

- Generator: `scripts/tune_routing.py`.
- Exact certificate: `certificates/routing-tuned.json`.
- Upstream patch: `patches/h46-nonadjacent-76.patch`.

Run `make verify`; `make tuned-note` rebuilds this PDF. Negative checks reject reuse of  $\beta = 1/2$ , the old layout bound, the guard endpoint, or a nonpositive final absorption gap. Finite tests and certificates do not independently prove the upstream algorithmic interfaces. Independent mathematical review remains open.